

Chapter 1

Reviving Indigenous Clay Pot Production for Students' Entrepreneurial and Vocational Inspiration

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ABSTRACT

This book chapter provides a comprehensive overview of indigenous clay pot production, including its definition, importance, and history. It explores how clay pot production is practised in different parts of Africa, such as Ghana, Tanzania, and Nigeria, and its significance in various cultures. The chapter examines the types of clay used in pot production, different techniques used, and traditional tools and equipment employed. Moreover, the chapter highlights the potential of clay pot production for students' entrepreneurship and vocational career development. It delves into the challenges of pursuing a career in clay pot production, such as the limited market for pottery products, access to raw materials, learning curve, physical demand, business challenges, and creative isolation among others. Through case studies, the chapter shows how clay pot production has aided entrepreneurial success despite these challenges. This chapter is a valuable resource for anyone interested in understanding the world of clay pot production and its potential for economic and cultural development.

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INTRODUCTION

Research has shown a growing concern among African scholars about the need to return to indigenous arts (Mugovhania, 2012; Ndubuisi, 2022; Zondi, 2021). The recognition of the gradual extinction of the history, knowledge, relevance, and practice of African traditional art and culture in modern African society may occasion this. For instance, it is argued that poor documentation of the rich and cherished art and cultural heritage of the African people has hindered the growth of African culture and the poor practice of indigenous art among Africans (Ekong, 2018). This cultural impoverishment affects indigenous art's historical, social and technological relevance and the economic and entrepreneurship opportunities associated with ancient African artefacts. It is also obvious that most of the works on African indigenous art and craft are merely artistic writing which only focuses on presenting to students their historical antecedents, styles, and techniques and neglects their economic potential (Elebute & Odokuma, 2016). Pottery is one of the indigenous arts that has been in practice for ages and is still relevant in modern society due to the wide usage of pots within traditional African settings. For instance, it is reported that clay wares are used for traditional religious rituals, traditional worship, cooking, eating, fetching and preserving water, and storing and preserving food items and clothes in ancient traditional Sub-Saharan African society (Busari & Odetoyinbo, 2021). Umoru-Oke (2017) argued that indigenous clay pots have innate functions in African culture which cannot be replaced by modern synthetic products such as rubber, enamel, and plastics and that indigenous pottery is an ongoing, dynamic and viable art used in many Yoruba communities in Nigeria to empower young girls in modern times. Contrary to the view of potters that indigenous clay pot art is no more financially rewarding due to modernity (Ayuba, 2009), one cannot deny the fact that clay pots still have many uses and functions in contemporary society. In the Niger Delta region of Nigeria, the 'evwere' is used to quickly thicken banga soup and speed up the cooking time for any soup, stew, or broth (Ugboma, 2021). Researchers have argued that cooking with indigenous clay pots, which are produced from natural materials, makes food taste better, healthier, and juicier than cooking with modern cookware because it almost preserves all the micronutrients present in the food (Khan & Banerjee, 2020; Saxena et al., 2021).

There are huge entrepreneurial opportunities for clay pot art in modern society. It is not enough to expose students to the historical perspective of indigenous arts and analyse their styles and techniques to promote their knowledge, practice and relevance in modern society. Students should be made to see entrepreneurial and economic opportunities in indigenous arts and develop indigenous vocational skills for self-reliance (Major & Leigha, 2019). Okonkwo (2014) reported that more than eighty (80) per cent of students interviewed had the perception that ceramics education is not economically viable and thus may not guarantee vocational opportunities in society. This may be because the indigenous traditional arts curricula do not provoke entrepreneurial and vocational inspirations among students but merely analyse the arts' antecedent, history, relevance and practice.

Therefore, this chapter aims to present a detailed process of traditional indigenous clay pot production from real-time studio experience and the vocational and entrepreneurial opportunities it can provide students in modern society. The chapter starts by defining indigenous clay pot production and explains the importance of indigenous clay pot production. It also presents a brief history of indigenous clay pot production, indigenous clay pot production in different parts of the world, uses of indigenous clay pots in various cultures, types of clay used in indigenous clay pot production, different techniques in indigenous clay pot production, traditional tools and equipment used in indigenous clay pot production

and the benefits of indigenous clay pot production to vocational career development. The chapter also presents the challenges faced in indigenous clay pot production, opportunities available in indigenous clay pot production, strategies to address challenges and take advantage of opportunities. Lastly, the chapter reviews some case studies on how indigenous clay pot production can spur entrepreneurship and vocational careers among students.

DEFINITION AND IMPORTANCE OF INDIGENOUS CLAY POT PRODUCTION

Indigenous clay pot production is an ancient traditional art commonly referred to as pottery (Asmah et al., 2013). It is also called Earthenware (Abbacan-Tuguic & Galinggan, 2016). Research has shown that indigenous clay pot production is an aspect of industrial art called “Ceramics art”, a branch of fine and applied arts concerned with producing wares from clay and subjecting them to heat treatment for permanence (Esosuakpo, 2020). Indigenous pot production is one of the earliest complex technologies involving changing clay into ceramic, known as pottery (Okwoche & Okonkwo, 2021). Indigenous clay pot production is described as a traditional pottery made by firing a fashioning plastic clay into an object of desirable shapes under a particular temperature to produce permanent pots used for different cultural, social and personal purposes (Okwoche & Okonkwo, 2021). It is also defined as the traditional or cultural art that involves changing clay into pots and earthenware in response to the people’s social, economic and technical demands based on the development of several different civilisations from ancient to modern society (Olasebikan, 2022). No wonder indigenous clay pot production has names according to cultural differences and the origin of the art. For instance, it is called Isan pottery by Ekiti people in Ekiti State, Nigeria (Olasebikan, 2022), Mambong Pottery in Malaysia (Tajul et al., 2011), Zulu pottery in South Africa (Olalere, 2019), Sirigu traditional pottery and Mpraeso pottery in Ghana (Kusimi et al., 2020; Navei, 2021), and *Okpasi Ikoror* pottery by Yala people in Cross River State, Nigeria, among others.

Scholars across the globe have widely documented the importance of indigenous clay pot production. The importance of this ancient traditional industrial art ranges from maintenance of traditions, cultural artifacts, and indigenous knowledge (Kaari & Ombaka, 2019; Olasebikan, 2022), providing opportunities for inculcating in younger generation of women the spirit of industry and traditional technology development (Esosuakpo, 2020; Olalere, 2019), providing cultural tourism opportunities for traditional communities (Asmah et al., 2016; Navei, 2021; Yussif et al., 2018), providing opportunities for rural women to contribute to the socio-economic development of their country (Busari & Odetoyinbo, 2021), providing alternative source of livelihood for marginalised population (Elebute & Odokuma, 2016), and provide entrepreneurship and vocational or career opportunities for rural people in modern society (Pal, 2021). This implies that indigenous clay or traditional pot production has sociocultural, socio-economic, vocational and entrepreneurship potentials in modern society. The importance of indigenous clay pots in Africa includes history, culture, symbolism, technology, education and art (Asante et al., 2015).

INDIGENOUS CLAY POT PRODUCTION IN DIFFERENT PARTS OF AFRICA

This section discusses the processes and procedures used in the production of indigenous clay pots in different African contexts.

Ghana

In Ghana, indigenous or traditional clay pot production at Mpraeso pottery has been adequately documented by Kusimi et al. (2020). The process follows a sequence of activities called “*chaîne opératoire*”. This process involves extraction, processing, forming, drying, firing, sorting or packing, and transportation or utilisation.

Extraction: Potters mined clay material from Nkawkaw, Oframase, or Amanfrom, while the “amodine” (local sand) is mined from Nkorkrokumah.

Procession: The clay material is purified to remove all debris, after which it is soaked in water for a day or two to form a wet clay slurry so that the potter can easily work with the cleaned clay material. The potter adds filtered local sand called amodine and mixes it properly to smoothen, soften, and make the clay material plastic to prevent cracks when firing. The mixed clay material is then purged, pinched, wedged, and kneaded until the desired texture is obtained (see Figure 1).

Forming: The potter places the kneaded clay on a piece of board and on a potter’s, wheel called *dua* made locally, rolls the kneaded clay into a rod-like form, and coils it into a circular shape. The potter then walks around the *dua* to smoothen and cut the edges of the clay while moulding the pot. The potter uses a piece of fabric called *ntomago* dipped in water to give the pot perfect smoothening (see Figure 2).

Drying: The sun-drying of the pots is done two times. The first time is immediately after moulding. The inside and the outside of the pots are further smoothed using *sabroba* to beat and flatten the base, *sikyie* to polish both the inside and outside, *kwaboba* to give further a fined polishing texture, *dadwe* to create designs, and finally, *ahwenie* and *sapo* to smooth the pots further and prepare them for the second sun-drying (see Figure3).

Firing, sorting/packing, and transportation/utilisation: Pots are open-fired on low-temperature in the backyards of the residence of the potters, using firewood and sawdust. The sawdust is meant to give

Figure 1. Soaking, adding amodine, and kneading of the mixed clay material



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Figure 2. The forming process in clay pots manufacturing



Figure 3. Drying and smoothening process of clay pots manufacturing



the pots a black colour. After the firing process, the pots are sorted out from the fire and are ready for sale (see Figure 4).

Tanzania

In Tanzania, the indigenous or traditional clay pot production is documented by Mteti (2016), specifically in Kisi and Pare of Southern Highland and North-East highland of Tanzania. In Tanzania, pottery art was originally meant for women. However, with the introduction of the potter's wheel, men started venturing into the vocation. Pot production has three processes (clay extraction and transportation, clay preparation, pot making, firing, decoration, and marketing) with sub-activities in each stage.

Clay extraction and transportation: The potters travel kilometres to extract clay materials from farm-lands, transport the clay materials to their homes, store the materials in their homes and preserve the

Figure 4. The firing and sorting processes in clay pots manufacturing process



materials either in bags or spray them on the ground and cover them with other substances to prevent theft (see Figure 5).

Clay preparation: The potters take time to remove impurities like pebbles, rocks or leaves from the extracted clay material. The clay is soaked in water and, at the same time, kneaded using hands while removing impurities such as stones and other unwanted substances. After proper mixing, the clay material is heaped at the potting area, which can be potter's wheels for modern technology or pieces of board placed on bare ground for traditional technique.

Pot making: The potters start moulding the pot by forming a lump of clay with a hole in the centre. The lump of clay is placed on a ring on a banana leaf for the traditional approach, while the modern approach is made on the potter's wheel. In the case of Tanzania, traditionally, pots are made in two halves (the top and the base), and then the top and base are joined together to form a pot. Scraping of the base is done until it is perfectly smoothened (see Figure 6).

Figure 5. Storage of clay materials at home



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Figure 6. Traditional and modern pot making in Tanzania



Pot firing: The potters in Kisi and Pare adopt an open firing technique just as those in Ghana under controlled temperatures. While Pare potters prefer to use small twigs of tree species called Mathanzu in Kipare to fire the pots, potters in Kisi use firewood from a tree known as ‘Nguti’ to fire theirs. It is reported that the modern firing approach adopted uses a kiln for black colouration (see Figure 7).

Decoration and marketing: This process is done after firing in modern pot production in Pare, while for Kisi, it is done before firing. The potters use both traditional designs and modern designs. In Pare, after firing the pots, they are immediately covered with leaves of a nkasha tree to give the pots a bright colour (see Figure 8). However, potters in Kisi decorate their pots at the end of the shaping process while the clay is still malleable. The finished products are moved directly to the consumers or distributed through a third party.

Figure 7. Firing of pots after moulding



Figure 8. Decoration of pots using leaves of a Nkasha tree



Nigeria

In Nigeria, indigenous clay pot production at Ebo Yala pottery in Cross River State has also been properly documented by Okwoche and Okonkwo (2021). The production techniques are the same as that of Ghana and Tanzania, even though the processes at Yala pottery are thoroughly reported. The process starts with clay collection, clay preparation, pot making,

Clay collection and storage: Traditionally, the soil texture meant for clay pods is known and called “Ome”. The ” Ome ” soil belongs to the clay soil family found in selected environments or farmland in the Yala kingdom, while some are collecting the brink of the river bank. A shovel, hoe, knife or spade is used to dig the clay. The collected clay material is taken home for the stage (see Figure 9). The clay material is stored either in bags or bowls to avoid hardening.

Clay preparation: Just as in other countries, the clay is prepared through cleaning, pulverisation, mixing, and kneading. At the cleaning stage, the potter uses bamboo or a shell to slice the clay and remove

Figure 9. Clay collection and storage in Yala



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the particles not needed in the clay, such as stone, water, dirt or decomposed materials. Immediately after this stage, the clay is pulverised (pound) with a wooden pestle and sieve. The sieved clay is then mixed with sand to make it plastic. After pulverising the clay, the kneading process starts. The potter can use his hand, wood or bamboo to knead the clay while adding dried husk gently until the desirable clay texture is obtained. The potter has to cover the kneaded clay with wet gumming bags to maintain its softness and wetness (see Figure 10).

Clay pot making: This stage involves forming or fashioning the pots. It can be done using hand-finger-forming or wheel-forming approaches. The potter places a sizable roll of clay and uses an anvil to beat it to an elongated flat mass to form the pot's base. Similarly, the potter uses the same approach to form the body (IbeyiIbeyi). When the pot base attains some hardness, the potter soaks it again in water to soften the surface. Then the thick part of the lower body is gently beaten with bamboo, and the anvil is held from inside with the right hand while rotating it on the concave base. The next stage involves forming the neck. The same pattern is used as both the shortened end of the circular plates are drawn together and are roundly pressed to fix with each other and transform into an undifferentiated mass by pressing with the fingertips. The last process here is smoothening. The potter uses an old local sponge made from a sack bag soaked inside the water to smooth the surface and then leaves them to dry under the sun. All these processes are presented in Figure 11).

Pot decoration and sun drying: Like the Kisi potters in Tanzania, Ebo Yala potters decorate their pots before drying and firing. It is reported that decoration uses an impression, roulettes, maize cob, bamboo, rope, and shell. The most popular ones are roulette (glazed) hand painting, slip painting, engraving and patterns (Okwoche & Okonkwo, 2021). After decorating or fashioning the pots, they are exposed to mild sunlight in the shade under controlled temperature for 2-3 days. This process is intentionally designed to protect the pots from damage due to excess heat (see Figure 12).

Pot firing: Firing and marketing are the last stages in the Yala pottery pot production process. Firing helps to transform the plastic clay into plastic. The Ebo Yala pottery firing process is called “bonfire firing”. They use specific firewood called “iginibe” with the help of mall twigs of the stick (ochi) in

Figure 10. Clay preparation by Yala potters



Figure 11. Clay making by Yala potters



Figure 12. Clay pot decoration and sun drying by Yala potters



Yala. To start the firing process, the firewood logs are lined up round on the ground with the clay pots placed on them, beginning from the middle until all the pots are arranged properly. The potters covered the arranged pots with banana leaves and grasses and then hit the fire (see Figure 13). The pots are taken to their local markets and sold to people from neighbouring communities.

Pot colouring, dying and marketing: After the firing process, the pots are set up for colouring and dying. The report states that Yala potters colour their pots using pigment extracted from Yede. They do this by dipping their fingers into the prepared colour or dye and drawing curved lines on the pots, including signs and symbols such as dots, flower drawings, and slashes. After the colouring and dying processes, the pots are taken to their local weekly markets and sold to people from Ogoja and other neighbouring communities (see Figure 14)

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Figure 13. Pot colouring, dying and marketing in Yala



Figure 14. Pot firing in Yala



IMPORTANCE OF INDIGENOUS CLAY POTS AND CLAY POTS PRODUCTION IN VARIOUS CULTURES

Research has indicated that in Kenya, indigenous clay pots are used for cooking, brickmaking, and plastering floors and walls and are major sources of livelihood for rural women (Kaari & Ombaka, 2019). The same source reported that many rural people in Kenya prefer to cook with indigenous clay pots because of the belief that the food cooked from clay pots tastes different compared to conventional utensils. In Kenya also, indigenous art is used as a national model of poverty reduction among rural women by empowering them through training and financial support (Ongachi et al., 2017). It is reported that rural farmers in Kenya prefer to store seeds in indigenous clay pots for incubation before planting because clay pots produce higher seed quality than those stored in polythene bags (Maina et al., 2017). In India, apart from the ceremonial use of clay pots for cooking food, storing water, and religious and smoking purposes, which are common to all cultures, indigenous clay pots were

also used as traditional tiles before the emergence of modern building practices (Panda et al., 2019; Rao & Lal, 2010). Another interesting use of indigenous clay pots is among the traditional Akans of Ghana, where these pots are produced in different special forms and used for serving food during Marriage Rites, Puberty Rites, Death and Burial rituals, and traditional rituals (Ancestral Veneration and Shrines) among others (Adjei et al., 2015).

To date, Kalinga people in the Philippines are still using indigenous clay pots for rice cooking (ittoyom), vegetable/meat cooking (oppaya), and water storage (immosso) pots (Abbacan-Tuguic & Galinggan, 2016). The Kalinga people believe that globular clay pots are far more advantageous than other organic containers because the heat has little or no effect on the containers and their contents, and they protect the food from moisture and pests. In Nigeria, water pot (àmù), cooking pot (ape idáná), money box (kóló), local stove (àdògán), flower pot (ìkòkò òdòdó), frying pot (àlâtí/ agbada), herbal pot (orù), local oil lamp (fitílà), plate (àwo ebo), soup pot (àsèje), placental pot (agbébi), sieve (ajere), and coal pot (kólùpótù) are still produced and used by the Yoruba people. Among the Akwa Ibom people, 'indigenous clays pots such as 'abang mmong' (water pot), 'Okò Ibíbio' (cooking pot), 'Usan Ibíbio' (bowls), 'Okò ata' (basin), 'etok okò' (small cooking pot), 'etok abang' (small water pot), 'oko ukot' (Palm wine pot) and 'abang isong' (big ceremonial pot for communal palm wine drinking) are still common (Peters, 2021). Like the rural Kenyan women, Akwa Ibom rural women in Nigeria believe that food cooked from indigenous pottery pots has a unique good taste which differs from food cooked from enamel or aluminium pots. The importance of indigenous clay pot production and the use of clay pots among different cultures is inexhaustive. In many cultural settings like the Philippines, pot making has been developed into modern usage and modified into ceramics for different modern uses.

TECHNIQUES USED IN INDIGENOUS CLAY POTS PRODUCTION

The techniques used for indigenous pot production are peculiar to different cultures. In Nigeria, the Yoruba people adopt three common techniques: direct, indirect and coiling moulding techniques (Fatuyi, 2018; Nanashaitu, 2017). When a potter uses the direct moulding technique, the pots are moulded without a pre-mould. The direct technique involves the following process: opening, coiling, consolidating the rim, expanding the belly, shaping the pot, scraping the pot and lastly, smoothening the pot. On the other hand, the indirect technique is used when the potter places the base of the pot upside down and involves rolling clay into a ball, making it into a flat surface, moulding on pre-mould, coiling, rim forming, smoothening and smoothening of the outer surface. Similarly, the coiling technique, known in Yoruba as 'ooran', involves building up the pot wall with superimposed rolls of clay coils until the desired shape of the pot is achieved (see Figure 15).

In Tanzania, two techniques are applied by potters: the traditional technique and the modern technique (Mteti, 2016). In the traditional technique, the lump of clay is placed on a ring placed on a banana leave, and the potter keeps using their hand to shape the lump of clay until the desired shape is formed. In this technique, pots are made in two halves (the top and the base), and the potter gradually adds clay to join the top and base to form a pot. In the modern technique, the potter applies the potter's wheel, where the lump of clay is placed on the wheel and gradually added to form desired pot (see Figure 16). This implies that techniques for making clay pots can be direct, indirect, coiling, traditional and modern.

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Figure 15. Direct, indirect and coiling techniques of making clay pots in Yoruba



Figure 16. Traditional and modern techniques of making clay pots in Tanzania



TRADITIONAL TOOLS AND EQUIPMENT USED IN INDIGENOUS CLAY POT PRODUCTION

Indigenous clay pot production is a traditional craft that has been practised for thousands of years around the world. The process of making clay pots involves various traditional tools and equipment. Traditional tools and equipment used in clay pot production are crucial in preserving cultural heritage, promoting sustainability, and enhancing artisanal skills. Using traditional tools and techniques, potters also reduce their environmental impact by using local and natural materials and minimising energy consumption. Additionally, the continued use of traditional pottery-making tools and techniques helps preserve tra-

ditional knowledge and skills passed down from generation to generation. This section discusses some traditional tools and equipment used in indigenous clay production.

- **Clay**

Clay is the main and most important raw material used in the traditional craft of clay pot production. It is a naturally occurring material composed of fine-grained minerals that can be moulded into various shapes when wet and hardened through firing. Clay is typically sourced from local deposits and is usually prepared by digging it out of the ground and refining it through washing, sieving, and drying. This process helps to remove any impurities, such as stones, organic matter, or debris, that might interfere with the quality of the clay. Once the clay has been refined, it can be used in pot production. The clay is typically mixed with water to form a workable paste that can be moulded into various shapes using a potter's wheel or hand tools. One of the most important properties of clay is its plasticity, which is the ability to be moulded and shaped without cracking or breaking (Bao et al., 2021; Tran Ba et al., 2016). This property makes clay ideal for making pottery, allowing potters to create various shapes and designs. Clay is also highly porous, which means that it can absorb water. This property is important in clay pot production because it allows the pots to “breathe,” which helps to regulate the temperature and humidity of the contents inside. The type of clay used depends on the region and the intended use of the pots. Once the pots have been shaped and decorated, they are fired in a kiln to harden the clay and make it durable.

- **Potter's Wheel**

The potter's wheel is a traditional tool and equipment for clay pot production. It shapes and moulds clay into various forms, including bowls, plates, cups, and vases. The potter sits at the wheel and uses their hands to shape the clay as it spins on the platform. Using the potter's wheel in clay pot production dates back thousands of years and has been an essential tool for potters worldwide (Méry et al., 2007). It allows them to create symmetrical and consistent shapes, which would be difficult to achieve by hand alone. The process of using the potter's wheel involves several steps. First, the potter takes a lump of clay and centres it on the wheel by pressing it firmly onto the centre of the platform. Then, using their hands, the potter applies pressure to the clay as it spins, shaping it into the desired form. The potter's wheel allows for a wide range of shapes and sizes to be created, from small cups to large vases (Berg, 2007). Once the pot has been shaped on the wheel, it is typically allowed to dry and harden before being fired in a kiln.

- **Clay Cutter**

A clay cutter is a traditional tool used in clay pot production designed to cut the clay into specific sizes and shapes for use in pottery making. Clay cutters are typically made of wire or thin metal blades attached to a handle or frame. A clay cutter is essential for preparing clay on the potter's wheel (Müller, 2007). After removing them from the wheel, the clay cutter is also used to trim and shape the edges of pottery pieces. There are different clay cutters, each designed for a specific purpose. The wire clay cutter, for example, is used to slice through blocks of clay to create even-sized pieces. On the other hand, the metal blade clay cutter is used to trim and shape the edges of pottery pieces. Using clay cutters in clay pot production is an important part of traditional craft. It

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allows potters to work with clay efficiently and effectively, ensuring that each piece of pottery is of a consistent size and shape. Without clay cutters, the process of pottery making would be more time-consuming and less precise.

- **Rolling Pin**

Using a rolling pin in clay pot production allows potters to create flat, even slabs of clay that can be used to make various pottery pieces (Triplett, 2000). While rolling pins are commonly associated with baking, they have also been used as a traditional tool and equipment in clay pot production. In pottery making, a rolling pin is typically used to flatten and shape slabs of clay into various forms (Peterson & Peterson, 2003). The rolling pin used in pottery is usually made of wood. It is designed to be sturdy and durable, with a smooth surface allowing easy clay rolling. For example, slabs of clay can be rolled into sheets and then cut and shaped into plates, bowls, or other forms. In addition to shaping the clay, a rolling pin can also be used to create textures or patterns on the surface of the clay. By rolling the clay over a textured surface, such as a piece of fabric or a carved block, the rolling pin can leave impressions that add visual interest to the finished pottery piece. While other tools, such as the potter's wheel and clay cutters, are also essential in pottery making, the rolling pin is a versatile tool that offers unique opportunities for creativity and expression.

- **Hand Tools**

Hand tools are essential to pottery making, and traditional clay pot production relies heavily on various hand tools. In addition to the potter's wheel, clay cutter, and rolling pin, various hand tools are used as traditional tools and equipment in clay pot production. These tools include spatulas, knives, and wooden ribs, among others. Spatulas are flat, thin tools used to smooth and shape the surface of the clay. They are especially useful in creating curves and contours on the surface of a pot. Spatulas come in various shapes and sizes, including straight, curved, and angled, to allow for a range of manipulation.

Knives are used to cut and shape the clay into specific forms. A sharp blade can be used to trim excess clay from a pot, while a serrated blade can be used to create texture on the surface of the clay. Knives come in various shapes and sizes and can be metal or plastic.

Wooden ribs are flat tools made of wood or plastic used to smooth and shape the surface of a pot. They are especially useful in creating curves and contours and can compress and smooth the clay as it is being shaped. Wooden ribs come in various shapes and sizes and can be used inside and outside a pot.

Wire and loop tools play a crucial role in creating clay pots. Wire tools are primarily used for cutting clay. A wire tool consists of two wooden handles with a wire between them, and it is used to cut through the clay. This tool is used to remove the clay from the potter's wheel after it has been shaped, and it can also be used to trim excess clay from the pot. Loop tools are used to shape the clay. A loop tool consists of a metal blade with a loop at one end and a wooden handle at the other. This tool scrapes clay from the pot's surface, creating different shapes and textures. Loop tools can also be used to smooth the clay pot's surface. By gently scraping the surface of the clay, the potter can create a smooth surface that is perfect for decorating. Both wire and loop tools can create fine details in the clay. For example, wire tools can create lines or grooves in the clay, while loop tools can create intricate patterns and designs.

- **Kiln**

A kiln is an essential traditional tool used in clay pot production. It is a furnace or oven for drying, baking, and firing clay objects at high temperatures. Kilns have been used for thousands of years to create pottery and ceramics (Bienkowski & Millard, 2010), and they have evolved, but the basic principles of their operation have remained the same. The kiln plays a critical role in shaping and transforming raw clay into functional and decorative objects in clay pot production (Zola et al., 2020). Before the clay pots are fired in the kiln, they are shaped by hand or with the help of a potter's wheel. The raw clay pots are left to dry in the open air or a shaded area until they are leather-hard. Once the pots are dry enough, they are loaded into the kiln, which is heated to temperatures ranging from 500°C to 1300°C (McClellan & Dorn, 2006). The firing process can take anywhere from a few hours to several days, depending on the size of the kiln and the thickness of the clay pots. The kiln's temperature gradually increases during the firing process, causing the clay pots to undergo physical and chemical changes. At around 600°C (Ashby, 2013; Wroe, 1994), the clay begins to transform into a hard and durable material known as ceramic. At higher temperatures, the clay vitrifies, meaning it becomes glass-like, and the glaze applied to the pots melts, fusing to the surface of the clay and creating a smooth, glossy finish.

- **Glazes**

Glaze is a traditional tool used in clay pot production to create a smooth and glossy surface on clay objects. It is a mixture of finely ground minerals and chemicals suspended in water, which is applied to the surface of the clay pot before firing in the kiln. The glaze provides a protective coating on the clay pot, preventing water from penetrating the surface and making it more durable (Maltoni et al., 2012). It also enhances the beauty of the clay pot by adding colour, texture, and lustre to the surface. Glazes come in various colours and textures; potters can create their custom glazes by mixing different minerals and chemicals in various proportions (Rice, 2015). Applying glaze to the clay pot is a crucial step in the pottery-making process. The potter must ensure that the glaze is applied evenly and smoothly to avoid bubbles, drips, and other imperfections that can affect the final appearance of the pot. Once the glaze is applied to the clay pot, it is fired in the kiln at high temperatures, causing the glaze to melt and fuse to the surface of the clay (Levy et al., 2022). This process creates a smooth, glossy finish and a layer of protection on the pot. In addition to enhancing the beauty and durability of the clay pot, a glaze can also be used to create specific effects on the surface, such as crackling, crystallisation, and speckling. These effects can be achieved by manipulating the glaze's composition and the kiln's firing temperature.

- **Decorating Tools**

Traditional clay pot production often involves various decorating tools that can be used to create unique and intricate designs on the surface of the pots. The most common tools used in pottery decoration include brushes, stamps, and carving tools. Brushes are commonly used in pottery decoration for painting designs onto the surface of the pots. These brushes are often made from natural materials like bamboo, animal hair, or plant fibres, which can provide a variety of textures and effects on the pot's surface. Different brush sizes and shapes can create fine details or larger patterns. Stamps are another common decorating tool in pottery production. These can be carved from wood, clay, or stone to create repeating patterns or designs on the pot's surface. Stamps can be used to create a variety of textures and

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shapes and can be used to create intricate designs quickly and easily. Carving tools create designs by carving into the pot's surface. These tools can include knives, chisels, and other sharp instruments that can be used to create intricate patterns and textures. Different tools, such as fine lines or deeper grooves, can create different effects. Overall, these tools are essential in traditional clay pot production and can help to create unique and intricate designs on the surface of the pots. The skilful use of these tools can result in beautiful and functional pottery that is not only aesthetically pleasing but also durable and long-lasting.

- **Drying Racks**

Drying racks are essential tools in traditional clay pot production, where they are used to dry newly formed pots before they are fired. In this process, freshly made clay pots are placed on the drying rack, allowing them to air dry evenly and gradually to prevent cracking and warping during firing (Peterson & Peterson, 2003). In traditional clay pot production, drying racks are typically made from natural materials such as bamboo or wooden poles, ropes, and netting. The racks are designed to be simple and efficient, allowing for good airflow around the pots to facilitate drying. The drying racks come in different sizes and shapes, depending on the size and shape of the pots being made. For instance, small drying racks may be used for small pots, while larger ones may be used for bigger pots. Some drying racks are also adjustable, allowing the size to be modified to fit different pots (Orton & Hughes, 2013). To use the drying rack, the freshly made pots are carefully placed on the racks to allow for adequate airflow around each pot. The racks are placed in a well-ventilated area to ensure the pots dry evenly.

POTENTIALS OF INDIGENOUS CLAY POT PRODUCTION FOR STUDENTS' ENTREPRENEURSHIP AND VOCATIONAL CAREER DEVELOPMENT

Indigenous clay pot production has a long and rich history, with various communities and cultures developing unique styles and techniques. There has been a renewed interest in traditional pottery recently, with many people seeking handmade and artisanal products. This has increased the demand for traditional clay pots, creating opportunities for individuals and communities to reap many benefits. One of the key benefits of traditional clay pot production is the development of vocational skills. To produce high-quality clay pots, individuals must learn various skills, including pottery throwing, glazing, and firing. These skills can be acquired through vocational or technical schools, apprenticeships, or self-directed learning. Vocational training programs can provide students with a solid foundation in the basics of pottery production, including techniques for shaping, trimming, and finishing clay pots.

Additionally, the demand for traditional pots for domestic and decorative purposes and for cooking and serving indigenous dishes in hotels and restaurants creates job opportunities (Adebimpe, 2015). Therefore, individuals can start their businesses and become self-employed, selling their pottery creations and contributing to the economic development of their communities. Starting a pottery business involves much more than just creating pottery. Individuals must learn to manage finances, develop a marketing strategy, and understand customer needs and preferences. These skills can be honed through practical experience and mentorship from experienced potters. Furthermore, entrepreneurial studies have been introduced into the Nigerian education system, equipping students with the skills and drive to establish successful enterprises in traditional industries such as pottery (Anthony-Euba & Towobola, 2014). These educated graduates are expected to introduce improved technologies to traditional indus-

tries, like pottery making, which would reduce failure rates and, in turn, positively impact the national economy. Furthermore, using better tools in traditional pottery making will result in less strenuous work for potters, enabling them to be more creative (Alimba & Mgbada, 2003).

Moreover, indigenous clay pot production can help preserve culture and heritage. Pottery making has been an essential part of many cultures for thousands of years, with unique styles, techniques, and designs that reflect the history and traditions of the community. By learning and practising this craft, individuals can help keep these traditions alive and pass them down to future generations, thus strengthening cultural identity and creating a sense of belonging within the community. In many indigenous communities, pottery-making has been a communal activity that brings people together and strengthens social bonds. By learning and practising this craft, individuals can connect with others in their community, build relationships, and create a shared identity (Nortey & Bodjawah, 2018). This can promote social inclusion and create a more vibrant and cohesive society.

The production of traditional clay pots can also promote sustainability and environmental stewardship. Traditional pottery techniques often use locally sourced materials and natural methods, which can promote sustainability and reduce the industry's environmental impact. By incorporating these practices into modern production methods, individuals can create more eco-friendly and sustainable pottery, thus contributing to a more environmentally responsible industry. As an example, a study has shown that the creation of artisanal products such as clay pots can impact changes in land use and cover, and the existence of traditional governance organizations can aid in establishing a link between producers and markets to encourage more enduring management of territorial resources like the tropical dry forest, which is ecologically valuable but frequently disregarded (Lira et al., 2022).

The tactile nature of pottery production can also help individuals develop their fine motor skills and hand-eye coordination. Pottery making is an art form, and individuals can express their creativity by designing and producing unique clay pots. The process of creating pottery can be therapeutic and meditative, providing a means of relaxation and stress relief. In addition to vocational and entrepreneurial skills, students pursuing a traditional clay pot production career can develop their creative expression. Pottery making is an art form, and students can express their creativity by designing and producing unique clay pots. Creating pottery can be therapeutic and meditative, providing relaxation and stress relief for students. Additionally, the tactile nature of pottery production can help students develop their fine motor skills and hand-eye coordination.

CHALLENGES ASSOCIATED WITH PURSUING A CAREER IN INDIGENOUS CLAY POT PRODUCTION

Despite the numerous benefits of traditional clay pot production, there are also challenges associated with pursuing a career in this field. It has been stated that potters are facing several challenges in their profession, including a lack of access to working capital and outdated equipment (Adebimpe, 2015; Ogunyoku et al., 2011). Other challenges that individuals interested in a career in clay pottery may face include the following:

- **Limited Market for Pottery Products**

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While there is a growing demand for handmade and artisanal products, the market for traditional clay pots is still relatively small. This means that individuals who want to make a living from producing pottery must be creative and innovative in marketing and selling their products. This may involve developing a strong online presence, targeting niche markets, or collaborating with other artists and designers to create unique and distinctive products.

- **Access to Raw Materials**

The availability and cost of clay and other materials used in pottery production can vary depending on the location. In some areas, individuals may be able to source materials locally, while in others, they may need to import materials from other regions or countries. This can also increase production costs and impact a pottery business's profitability.

- **Learning Curve**

One of the most significant challenges of pursuing a career in clay pottery is the steep learning curve. It takes time and effort to master the art of clay pottery. A potter must learn to understand the properties of clay and how to use various tools, techniques, and firing methods to create functional and decorative pottery. This requires significant practice, experimentation, and patience.

- **Physical Demands**

Working with clay pottery is physically demanding. A potter must be able to knead, shape and mould clay for extended periods. It can be strenuous and requires stamina and good physical health.

- **Business Challenges**

The business side of pottery making is also challenging. Potters have to manage the cost of materials, production, marketing, and sales. Building a successful pottery business requires financial management skills and balancing creative expression with commercial viability.

- **Creative Isolation**

Pottery making can be a solitary activity that can be challenging for some people. Potters need to spend significant time alone, working on their craft. This can be isolating and require self-motivation and discipline to maintain focus and productivity.

- **Environmental Concerns**

Pottery making can also pose environmental challenges, especially during the kilning process, which usually involves heating moulded objects in a local furnace. Nearby grasses, microorganisms or animals useful to humanity are killed due to heat from the furnace. Potters need to be mindful of the environmental impact of their materials and production methods. They must choose sustainable materials and firing methods to minimise their carbon footprint and reduce waste.

- **Competition**

The pottery market can be highly competitive. Potters must thus be able to create unique and compelling designs to stand out in a crowded market, and they must also be able to market their products effectively to reach potential customers.

- **Financial Stability**

Like many creative professions, pursuing a career in pottery can be financially unstable. Potters may experience financial instability due to the seasonal nature of their work and the ups and downs of the market. It may be necessary to have a secondary source of income to supplement their pottery income.

CASE STUDIES ON HOW INDIGENOUS CLAY POT PRODUCTION HAS AIDED THE ENTREPRENEURIAL SUCCESS

Despite these challenges, there are many success stories of students who have pursued careers in indigenous or traditional clay pot production. For example, in India, the Khurja Pottery Cluster in Uttar Pradesh has become a hub for traditional clay pot production, with over 200 small and medium-sized enterprises producing various pottery products (Craft Ghar, n.d). Government initiatives have supported the cluster to promote traditional crafts and provide local artisans with training and business development services.

Another case study was presented by Yadav and Goyal (2015) concerning Mansukhbhai Prajapati, whose family was involved in traditional clay pottery in rural India. Mansukhbhai Prajapati, from rural India, dropped out of school and returned to his family's traditional business of clay pottery after facing setbacks in his education and career. While working at a brick roof tiles factory, Prajapati developed a machine to create more efficient clay products. He experimented with different types of clay to create innovative products like the Mitticool clay refrigerator, nonstick clay tawa, and clay pressure cooker. The Mitticool clay refrigerator gained attention for providing a low-cost alternative to traditional refrigeration by using natural evaporation to keep food and drinks cool without electricity. This made it ideal for rural areas with unreliable or unavailable electricity. Prajapati's success improved his financial situation and positively impacted his community, and he also received several awards, including recognition from the President of India, and inspired many seeking to impact their communities positively.

Between 2003 and 2013, the Sri Lanka project collaborated with Ashram International Charity in the UK to empower women through craft revival (Anjum, 2013). Artisanal potters worked with a modern ceramic factory to produce items for export, using traditional making methods such as throwing on the wheel or beating. In contrast, the factory provided glazing, firing, packaging, and shipping to Europe. The project adhered to fair trade principles, paying potters 30% more for their work than they would locally (Anjum, 2013). The project positively impacted craft revival, women's empowerment, lifestyle, and health and safety at the pottery. One of the ranges produced by the potters, called Matti, won the House Beautiful Gold Award for Best Tableware range in 2011 and was sold by retailer Jamie at Home.

CONCLUSION AND RECOMMENDATIONS

In conclusion, this chapter has provided valuable insights into indigenous clay pot production. It has highlighted the importance of indigenous clay pot production in various cultures, examined the different techniques and traditional tools used in the process, and explored the potential of pottery as a source of entrepreneurship and vocational career development for students. Moreover, the chapter has addressed the challenges of pursuing a career in indigenous clay pot production and provided case studies of successful entrepreneurs. These case studies demonstrate that despite the challenges, indigenous clay pot production can be a viable source of economic development in African countries and other regions where pottery is practised. Overall, this chapter has significant implications for students, scholars, and pottery enthusiasts interested in understanding the world of indigenous clay pot production. It highlights the potential of pottery as a source of economic and cultural development and underscores the importance of preserving traditional techniques and knowledge in the field. The chapter provides a foundation for future research and scholarship on pottery, entrepreneurship, and vocational career development in African countries and beyond.

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KEY TERMS AND DEFINITIONS

Entrepreneurship: In the context of this chapter, it refers to the act of starting a business or a series of businesses with the goal of creating a benefit to humankind, indigenous communities and society at large, and it focuses on helping communities or the environment through their products and services. It is one of the resources economists categorize as integral to production and the other three being land or natural resources, labor and capital.

Indigenous Arts: It is an expression of Indigenous people's identity, culture, spirituality and relationships to their indigenous communities and countries.

Indigenous Clay Pot Production: It is described as a traditional pottery made by firing a fashioning plastic clay into an object of desirable shapes under a particular temperature to produce permanent pots used for different cultural, social and personal purposes.

Indigenous Pottery: It is made by forming plastic clay into objects of required shapes and firing them to appreciable temperatures in the open or in pits to induce reactions that lead to permanent changes, including increase in strength, hardening and setting their shape.

Traditional Tools: Refers to the equipment, instruments, or devices that have been used historically or traditionally in a particular craft or trade, such as the tools used in indigenous clay pot production.